(Please write your Roll No. immediately) Roll No. .....

MID TERM EXAMINATION

EIGHTH SEMESTER (B. TECH), April 2024

Paper code: ETIT-402 Subject: Mobile Computing
Time: 1 Hour 30 Minutes Maximum Marks: 30

Note: Attempt three questions only.

1. **Question No.1 is compulsory.**
2. **Attempt any two questions from the remaining three questions.**

Q-1. Explain each of the following in short answer: (5 x 2)

(a) How do hard handoff and soft handoff differ in mobile cellular networks?

(b) Outline the characteristics of different generations of cellular systems?

(c) Using diagrams, differentiate between hidden terminals and exposed terminals in WLAN systems.

(d) Enumerate the fundamental features of a CDMA system.

(e) What are the primary functions of the Home Location Register (HLR) and Visitor Location Register (VLR) in mobile networks?

Q-2. (a) Evaluate the effectiveness of the three-tier architecture in mobile computing for ensuring scalability and flexibility in mobile applications. Provide examples to support your analysis. (5)

(b) Critically assess the role of WAP (Wireless Application Protocol) in facilitating mobile application development and deployment. Discuss its strengths and limitations, considering factors such as security, protocol stack, and application environment. (5)

Q-3.

(a) Evaluate the suitability of Bluetooth technology for various user scenarios outlined in the IEEE 802.15 standard. Consider factors such as data transfer rates, power consumption, and compatibility with different devices. (5)

(b)Compare and contrast the multiple access techniques used in wireless LANs, such as collision avoidance, polling, and spread spectrum. (5)

Q-4. Write a short note on **any two** of the following. (4 x 2.5)

a) What are some common user scenarios for Bluetooth as defined by the IEEE 802.15 standard?
b) Describe the physical layer specifications for Bluetooth technology.
c) What functionalities are provided by the MAC layer in Bluetooth communication?
d) How is link management handled in Bluetooth networks?

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